National University of Computer & Emerging Sciences

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“UNO Card Game”

Project Proposal

Object Oriented Programming

Section: BCY-2A

Group Members:

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UNO Card Game Simulation in C++

1. Introduction

Background:

UNO is a popular card game that involves strategic play and turn-based mechanics. The game is ideal for implementing Object-Oriented Programming (OOP) concepts such as inheritance, polymorphism, and composition. This project will simulate an UNO game using C++ and showcase the practical application of OOP.

Problem Statement:

The project will address how OOP concepts can be used to design a card game efficiently, focusing on aspects like class hierarchy, data encapsulation, and interaction between objects.

Objectives:

- Implement an UNO card game simulation using C++.
- Demonstrate the use of OOP principles such as inheritance, encapsulation, and polymorphism.
- Develop a turn-based system where players draw and play cards according to UNO rules.
- Provide an interactive and GUI-based game experience

2. Scope of the Project

Inclusions:

The project will be designed and programmed on Visual Studio Code and will include GUI (Graphical User Interface). Some functions that will be implemented in the project include:

- Implementation of core UNO rules (draw, play, reverse, skip, draw two, wild cards, etc.).
- Classes representing different components: Game, Players, Deck, and Card types.
- Basic game loop allowing two players to take turns.
- Randomized deck shuffling and card distribution.
- Handling special cards (Reverse, Skip, Draw Two, Wild, Draw Four).

Exclusions:

- Multiplayer functionality beyond two players.
- Advanced AI opponents (players will be manually controlled).
- Network or online play features.

3. Project Description

Overview:

The project will involve designing an UNO card game using OOP principles in C++. The game will include classes representing the game structure and mechanics, with interactions between players and cards. Some classes included in the project would be Game, Deck, Player1, Player2, 4 classes for the different colors, etc. The turn-based gameplay will allow players to draw and play cards according to standard UNO rules.

Technical Requirements:

- Programming Language: C++
- IDE: Microsoft Visual Studio / Dev-C++
- Libraries: Iostream, string, math, etc.
- GUI

Project Phases:

1. Research and Planning: Understanding UNO rules and defining class structures by observing a real-life example.
2. Design: Creating class diagrams and establishing relationships between the different objects.
3. Implementation: Writing and testing core functionalities of the game.
4. Testing & Debugging: Ensuring game mechanics, such as drawing a number or colored card or drawing a power card, work correctly.

4. Methodology

Approach:

- Research and planning (UNO rules and OOP design)
- Coding (Game logic, deck management, and player turns)
- Creating separate classes for decks and players
- Managing game flow
- Implementation of special cards logic
- Testing & debugging (Identifying and fixing errors, improving gameplay experience)

Team Responsibilities:

- Member 1 is responsible for keeping track of the 2 Players' classes, the array of 5/7 cards they have, and the card they would draw.
- Member 2 would design the different colored classes and the power card classes.
- Member 3 would keep track of the Game and Deck of Card classes, making sure that inheritance and association are implemented properly.

5. Expected Outcomes

• <u>Deliverables</u>:

- A working C++ program that simulates an UNO game.
- An instruction manual for the players before the start of the game.

<u>Relevance</u>:
This project demonstrates key ICT and software development concepts such as:

- Object-oriented programming (inheritance, encapsulation, polymorphism, composition).
- Data organization through card decks and player management.
- Turn-based game mechanics implementation in C++.

6. Resources Needed

<u>Software</u>:

- Microsoft Visual Studio / Dev-C++
- GitHub for version control

<u>Other Resources</u>

- Online and real-world references for UNO rules and C++ OOP tutorials.
- Assistance from the instructor and lab assistant for debugging and implementation guidance.